EXPERIMENT – 9

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AIM:

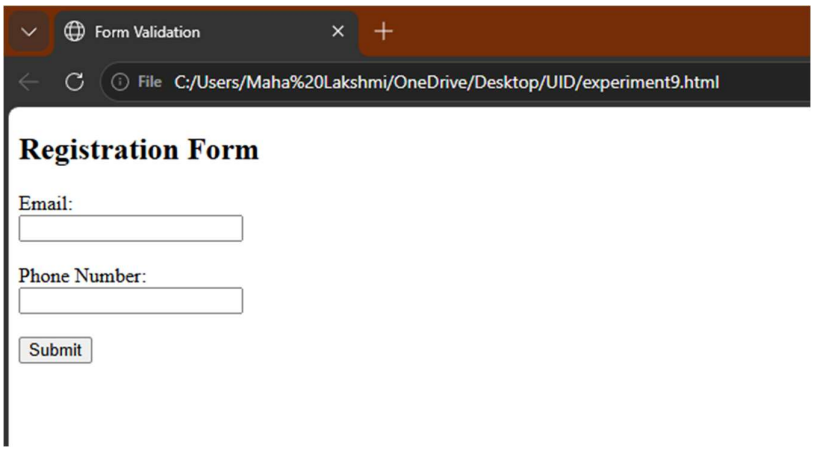
To design input forms that validate user data such as email and phone number and display error messages using HTML, CSS, and JavaScript with Validator.js.

Setting Up the HTML Form:

Program:

```
<!DOCTYPE html>
<html>
<head>
   <title>Form Validation</title>
</head>
<body>
<h2>Registration Form</h2>
<form>
   <label>Email:</label><br>
   <input type="text"><br><br>
   <label>Phone Number:</label><br>
   <input type="text"><br><br>
   <button type="submit">Submit</button>
</form>
</body>
</html>
```

Output:



The basic HTML form is displayed on the screen. It contains input fields for entering the email address and phone number, along with a submit button. No validation or styling is applied at this stage.

In this step, a simple form structure is created using HTML. The form includes input fields for user data collection and serves as the foundation for further styling and validation.

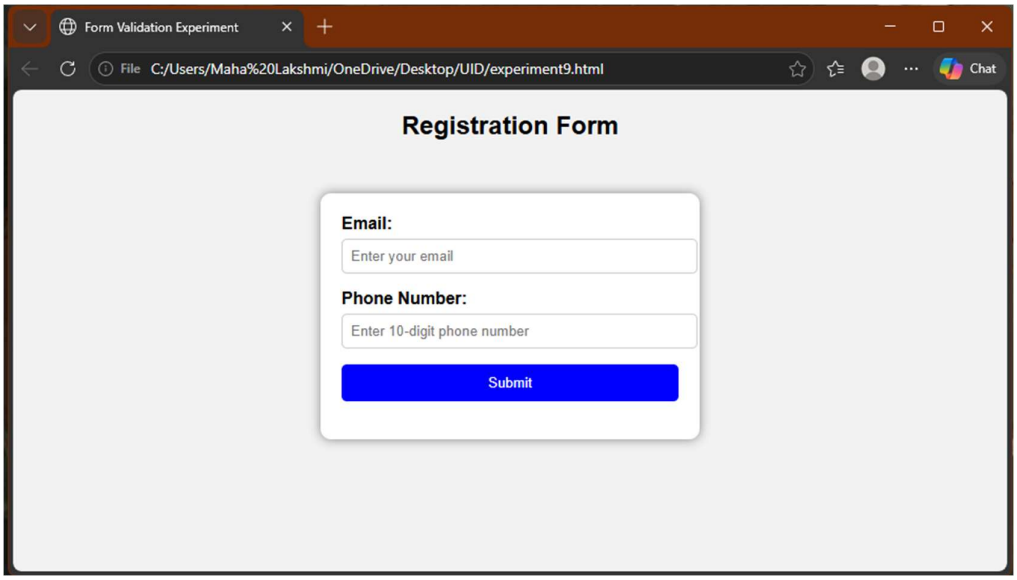
Styling the Form using CSS:

Program:

```
<style>
body {
    font-family: Arial;
    background-color: #f2f2f2;
}
form {
    background: white;
    padding: 20px;
    width: 300px;
```

```
    margin: auto;
    border-radius: 10px;
}
input {
    width: 100%;
    padding: 8px;
}
button {
    background-color: blue;
    color: white;
    padding: 10px;
    border: none;
    width: 100%;
}
</style>
```

Output:



The form is displayed with improved visual appearance. It is centered on the screen with a styled layout, including proper spacing, background color, and a formatted submit button.

CSS is applied to enhance the appearance of the form. Styling improves readability and user experience by organizing the layout and making the interface visually appealing.

Adding JavaScript Validation using Validator.js:

For Validator.js:

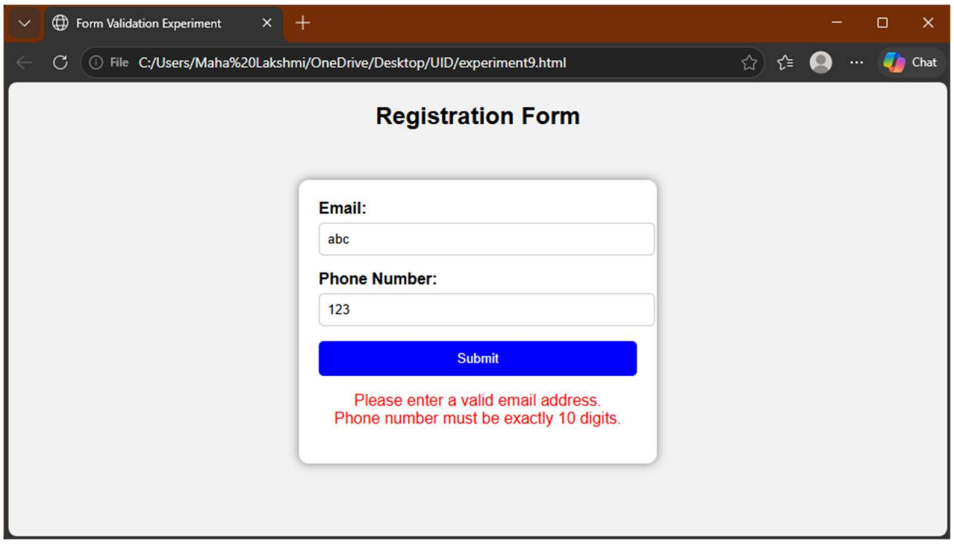
```
<script
src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.7.0/validator.min.js"></
script>


<script>
document.querySelector("form").addEventListener("submit", function(event) {
    let email = document.querySelectorAll("input")[0].value.trim();
    let phone = document.querySelectorAll("input")[1].value.trim();
    let error = "";
    if (!validator.isEmail(email)) {
        error += "Invalid email format.<br>";
    }
    if (!validator.isMobilePhone(phone, 'en-IN')) {
        error += "Invalid phone number.<br>";
    }
    if (error !== "") {
        event.preventDefault();
        document.body.innerHTML += "<p style='color:red'>" + error + "</p>";
    }
```

});

</script>

Output:



When invalid data is entered, appropriate error messages are displayed on the screen. Incorrect email format or invalid phone number triggers validation errors. When valid data is entered, the form is submitted successfully.

JavaScript along with Validator.js is used to validate the input fields. The isEmail() and isMobilePhone() functions ensure that the entered data follows the correct format. Error messages are shown for invalid inputs.

RESULT:

The input form was successfully created and validated using HTML, CSS, and JavaScript. The Validator.js library was used to check the correctness of email and phone number inputs, and error messages were displayed for invalid data.

CONCLUSION:

This experiment demonstrates how form validation improves data accuracy and user experience. Using Validator.js simplifies the validation process and ensures reliable input handling.